NANYANG TECHNOLOGICAL UNIVERSITY

SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/2024 Semester 1 MH1100 Calculus I Tutorial 11, Week 12

Your tutor will aim to discuss: Problem 1, 2, 3, 4, 5, 6, 7, and 8

Problem 1 Find $f(x)$
(a)
$$f''(x) = x^6 - 4x^4 + x + 1$$
(b)
$$f'''(x) = \cos x,\ f(0) = 1,\ f'(0) = 2,\ f''(0) = 3$$

[Solution:]
(a) We have
$$f''(x) = x^6 - 4x^4 + x + 1 \Rightarrow f'(x) = \frac{x^7}{7} - \frac{4x^5}{5} + \frac{x^2}{2} + x + C$$
Here C is a constant value. Further, we have
$$f'(x) = \frac{x^7}{7} - \frac{4x^5}{5} + \frac{x^2}{2} + x + C \Rightarrow f(x) = \frac{x^8}{56} - \frac{2x^6}{15} + \frac{x^3}{6} + \frac{x^2}{2} + Cx + D$$
Where D is a constant value.
(b) We have
$$f'''(x) = \cos x \Rightarrow f''(x) = \sin x + C \Rightarrow f''(0) = C$$
Since $f''(0) = 3$, we have $f''(x) = \sin x + 3$. Further
$$f''(x) = \sin x + 3 \Rightarrow f'(x) = -\cos x + 3x + D \Rightarrow f'(0) = -1 + D$$
Since $f'(0) = 2$, we have $f'(x) = -\cos x + 3x + 3$. Further
$$f'(x) = -\cos x + 3x + 3 \Rightarrow f(x) = -\sin x + \frac{3x^2}{2} + 3x + E \Rightarrow f(0) = E$$
Since $f(0) = 1$, we have
$$f(x) = -\sin x + \frac{3x^2}{2} + 3x + 1$$

Problem 2 A particle is moving with the the given conditions below. Find the position $s(t)$ of the particle
(a)
$$v(t) = t^2 - 3\sqrt{t},\ s(4) = 8$$
(b)
$$a(t) = 10\sin t + 3\cos t,\ s(0) = 0,\ s(2\pi) = 12$$

[Solution:]
(a) Since $s'(t) = v(t)$, we have
$$v(t) = t^2 - 3\sqrt{t} \Rightarrow s(t) = \frac{t^3}{3} - 2t^{\frac{3}{2}} + C \Rightarrow s(4) = \frac{64}{3} - 16 + C$$

Since $s(4) = 8$, we have $C = \frac{8}{3}$. And

$$s(t) = \frac{t^3}{3} - 2t^{\frac{3}{2}} + \frac{8}{3}$$

(b) Since $v'(t) = a(t)$, we have

$$a(t) = v'(t) = 10\sin t + 3\cos t \Rightarrow v(t) = -10\cos t + 3\sin t + C$$

Since $s'(t) = v(t)$, we have

$$v(t) = -10\cos t + 3\sin t + C \Rightarrow s(t) = -10\sin t - 3\cos t + Ct + D$$

Since $s(0) = -3 + D = 0$ and $s(2\pi) = -3 + 2\pi C + D = 12$, we have $D = 3$ and $C = \frac{6}{\pi}$. And

$$s(t) = -10\sin t - 3\cos t + \frac{6t}{\pi} + 3$$

Problem 3 Apply Newton's method to the equation $x^2 - a = 0$ to derive the following square-root algorithm (used by ancient Babylonians to compute $\sqrt{a}$):

$$x_{n+1} = \frac{1}{2}(x_n + \frac{a}{x_n}).$$

[Solution:]
We let $f(x) = x^2 - a$, so $f'(x) = 2x$. Use Newton's formula, we have

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - a}{2x_n} = \frac{1}{2}(x_n + \frac{a}{x_n})$$

Problem 4 Explain why we can not use Newton's method to find the root of the equation

$$x^3 - 3x + 6 = 0$$

if we choose $x_1 = 1$.

[Solution:]
The Newton's method is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

To find the root of the equation, we should let $f(x) = x^3 - 3x + 6$. In this way, the derivative is $f'(x) = 3x^2 - 3$. However, if we use $x_1 = 1$ as the initial value, we have $f'(x_1 = 1) = 0$. In Newton's formula, the derivative can not be zero.

Problem 5 What constant acceleration is required to increase the speed of a car from 50 km/h to 80 km/h in 5 seconds?

[Solution:]

We assume $a(t) = C\ km/h^2$. Therefore we have

$$v(t) = Ct + D$$

We assume at t_1, the speed is $v(t_1) = Ct_1 + D = 50\ (km/h)$, and at t_2, the speed is $v(t_2) = Ct_2 + D = 80\ (km/h)$.

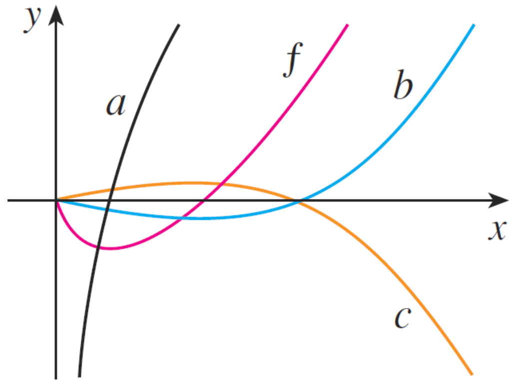
Since $t_2 - t_1 = 5\ (sec) = \frac{5}{3600}\ (h) = \frac{1}{720}\ (h)$. We have

$$v(t_2) - v(t_1) = Ct_2 + D - (Ct_1 + D) = 80 - 50$$

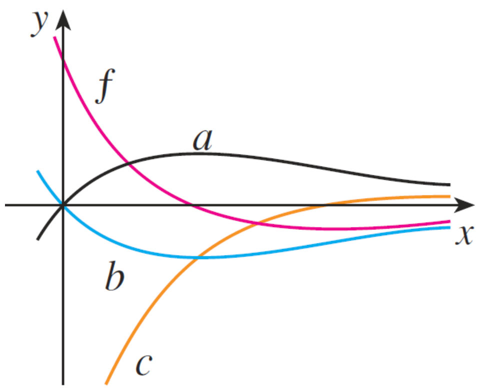
$$\Rightarrow C(t_2 - t_1) = 30 \Rightarrow C\frac{1}{720} = 30 \Rightarrow C = 21600\ (km/h^2)$$

Problem 6 Which graph is the antiderivative of f and why?

(a)



(b)



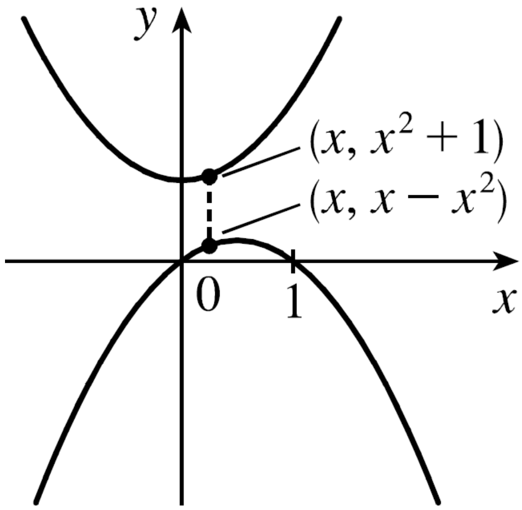
[Solution:]

(a) Curve b is the antiderivative of f. When f is negative, the antiderivative function should decrease. When f is positive, the antiderivative function should increase. The antiderivative function reaches the minimum at $f = 0$.

(b) Curve a is the antiderivative of f. Same reason as above.

Problem 7 What is the minimum vertical distance between the parabolas $y = x^2 + 1$ and $y = x - x^2$.

[Solution:]
We can plot the two parabolas as the figure below.



Therefore, the vertical distance $h(x)$ can be expressed as,

$$h(x) = x^2 + 1 - (x - x^2) = 2x^2 - x + 1$$

so we have

$$h'(x) = 4x - 1$$

and $x = 1/4$ is the only critical number. Since we have $h'(x) < 0$ for $x < 1/4$ and $h'(x) > 0$ for $x > 1/4$, at $x = 1/4$ the vertical distance $h(x)$ achieves the absolute minimum $h(x = \frac{1}{4}) = \frac{7}{8}$.

Problem 8 If 1200 cm^2 of material is available to make a box with a square base and an open top, find the largest possible volume of the box.

[Solution:]
Let b be the length of the base of the box and h the height. The surface area is

$$1200 = b^2 + 4hb \Rightarrow h = \frac{1200 - b^2}{4b}$$

The volume is

$$V = b^2 h = b^2 \frac{1200 - b^2}{4b} = 300b - \frac{b^3}{4} \Rightarrow V'(b) = 300 - \frac{3}{4}b^2$$

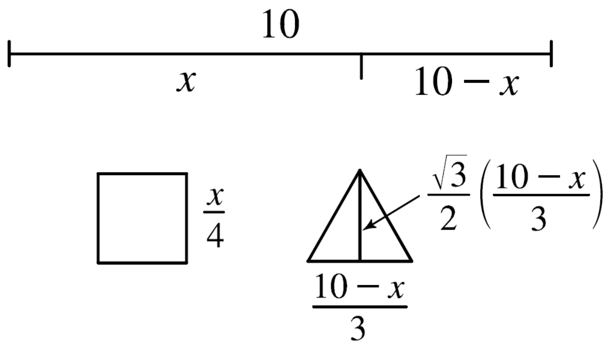
The critical number is

$$V'(b) = 0 \Rightarrow b = 20 \ (Ignore\ the\ negative\ value)$$

Since $V'(b) > 0$ for $0 < b < 20$ and $V'(b) < 0$ for $b > 20$, there is an absolute maximum when $b = 20$. In this way, at $b = 20$, the volume $V(b)$ obtains the absolute maximum value $V(b = 20) = 300 \cdot 20 - 20^3/4 = 4000$ cm^3, and the height is $h = \frac{1200 - 20^2}{4 \cdot 20} = 10$ cm.

Problem 9 A piece of wire 10 m long is cut into two pieces. One piece is bent into a square and the other is bent into a equilateral triangle. How should the wire be cut so that the total area is (a) a maximum. (b) a minimum.

[Solution:]
We can draw a graph as following,



We let x (m) be the length of the wire used for the square. The edge of the triangle is $\frac{10-x}{3}$ m, and the height of the triangle is $\frac{\sqrt{3}}{2}\frac{10-x}{3}$ m. In this way, the totally area is

$$A(x) = (\frac{x}{4})^2 + \frac{1}{2}(\frac{10-x}{3})\frac{\sqrt{3}}{2}(\frac{10-x}{3}) = \frac{x^2}{16} + \frac{\sqrt{3}}{36}(10-x)^2$$

Note that $0 \leq x \leq 10$.

Further we have

$$A'(x) = \frac{x}{8} - \frac{\sqrt{3}}{18}(10-x) = 0 \Rightarrow x = \frac{40\sqrt{3}}{9+4\sqrt{3}}$$

Further we have

$$A(0) = \frac{\sqrt{3}}{36}100 \approx 4.81, \;\; A(10) = \frac{100}{16} = 6.25, \;\; A(\frac{40\sqrt{3}}{9+4\sqrt{3}}) \approx 2.72$$

so (a) the maximum area occurs when $x = 10$ m, that is the whole wire is used for the square. And (b) the minimum area occurs when $x = \frac{40\sqrt{3}}{9+4\sqrt{3}} \approx 4.35$ m, that is use 4.35 m to for the square and the rest for triangle.

Problem 10 Show that the shortest distance from the point (x_1, y_1) to the straight line $Ax + By + C = 0$ is

$$\frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

[Solution:]

If $B = 0$, the line is just $x = -\frac{C}{A}$. And the distance is

$$|x_1 + \frac{C}{A}| = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

(Note if $A = 0$ in this situation, then C is also 0 and no straight line is available. So A can not be zero.)

If $B \neq 0$. We can rewrite the line function

$$Ax + By + C = 0 \Rightarrow y = -\frac{Ax}{B} - \frac{C}{B}$$

The square distance from (x_1, y_1) to the line is

$$f = (x - x_1)^2 + (y - y_1)^2 \Rightarrow f(x) = (x - x_1)^2 + (-\frac{Ax}{B} - \frac{C}{B} - y_1)^2$$

$$\Rightarrow f'(x) = 2(x - x_1) + 2(-\frac{Ax}{B} - \frac{C}{B} - y_1)(-\frac{A}{B})$$

$$f'(x) = 0 \Rightarrow x = \frac{B^2x_1 - ABy_1 - AC}{A^2 + B^2}$$

Further we have

$$f''(x) = 2(1 + \frac{A^2}{B^2}) > 0$$

In this way $x = \frac{B^2x_1 - ABy_1 - AC}{A^2+B^2}$ is the local minimum of the distance function. Actually, it is also the global minimum. At this point, the distance square is

$$f(x) = \frac{(Ax_1 + By_1 + C)^2}{A^2 + B^2}$$

Thus the smallest distance is

$$\sqrt{f(x)} = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$